

RETAIL SALES DATA ENGINEERING PIPELINE & BUSINESS INSIGHTS DASHBOARD

Submitted By

Keerthi JK (1CR22CS080)

CMR Institute of Technology

1. Executive Summary

This project implements an end-to-end Retail Sales Data Engineering Pipeline using Python and Power BI. The objective of the project is to ingest retail transaction data from Excel files, perform data validation, cleansing, transformation, enrichment, and generate an analytics-ready dataset that supports business intelligence reporting.

The solution follows a Medallion Architecture approach consisting of Bronze, Silver, and Gold layers to ensure data quality, governance, and scalability.

The final curated dataset is visualized using Power BI to provide insights into revenue performance, category trends, city-wise sales, profitability, and overall business performance.

2. Problem Statement

ABC Retail Solutions receives retail transaction data from multiple operational systems.

The data contains several quality issues such as:

- Duplicate transactions
- Missing values
- Invalid quantity values
- Invalid price values
- Inconsistent product naming conventions
- Inconsistent city names
- Different date formats
- Sensitive customer information

These issues make it difficult to generate accurate business reports and KPIs.

The objective is to build a complete data engineering pipeline that transforms raw data into a trusted analytics-ready dataset suitable for business reporting.

3. Project Objectives

The project aims to:

1. Ingest retail transaction data from Excel files.
 2. Perform data validation and quality checks.
 3. Clean and standardize the data.
 4. Apply data governance practices through PII masking.
 5. Enrich transactional data using product master data.
 6. Generate business KPIs.
 7. Build an interactive Power BI dashboard.
 8. Deliver actionable business insights.
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4. Solution Architecture

Architecture Diagram

Raw Excel Files



Bronze Layer (Raw Data Storage)



Validation Layer



Silver Layer (Cleaned Data)



PII Masking



Gold Layer (Business Dataset)



Power BI Dashboard

Architecture Components

Source Layer

Excel files containing retail transactions and product metadata.

Bronze Layer

Stores ingested raw data without modifications.

Validation Layer

Performs schema checks and data quality validation.

Silver Layer

Applies data cleaning and standardization.

Governance Layer

Protects sensitive customer information.

Gold Layer

Generates business metrics and reporting datasets.

Visualization Layer

Power BI dashboard for analytics and reporting.

5. Dataset Description

retail_data1

Contains retail transaction records collected from source system 1.

Attributes include:

- Transaction ID
- Customer Information
- Product Information
- Quantity
- Price
- Discount
- Payment Information
- Purchase Location
- Transaction Date

retail_data2

Contains retail transaction records collected from source system 2.

Structure is similar to retail_data1.

product_details

Contains standardized reference data.

Fields include:

- Product ID
- Product Name
- Category
- Standard Product Price

This table is used for data enrichment and validation.

6. Project Folder Structure

Retail-Sales-Data-Engineering-Pipeline

```
├── Documentation
├── Power BI
├── data
│   ├── raw
│   ├── bronze
│   ├── silver
│   └── gold
├── src
│   ├── ingestion
│   ├── validation
│   └── transformation
├── utils
├── main.py
├── kpi.py
└── README.md
```

7. Data Engineering Workflow

Step 1: Data Ingestion

Purpose:

Read source Excel files and ingest data into the pipeline.

Activities:

- Read Excel sheets using Pandas
- Load transactional datasets
- Load product reference dataset
- Store raw records

Output:

data/bronze/retail_raw.csv

Step 2: Data Validation

Purpose:

Ensure incoming data meets expected standards.

Validation Checks:

- Required columns exist
- Correct data types
- Missing value inspection
- Product ID consistency
- Duplicate record identification

Benefits:

- Improves reliability
 - Prevents downstream errors
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Step 3: Data Cleaning

Purpose:

Create a trusted dataset for analysis.

Cleaning Activities:

Duplicate Removal

Duplicate transactions are identified and removed.

Invalid Record Removal

Records with invalid prices or quantities are excluded.

Failed Transaction Removal

Only successful transactions are retained.

Standardization

Standardized:

- Product Names
- Categories
- City Names

Date Formatting

Different date formats are converted into a consistent structure.

Output:

data/silver/retail_cleaned.csv

8. Data Governance and Security

Customer information contains Personally Identifiable Information (PII).

To comply with privacy requirements:

Email Protection

Emails are hashed using SHA256.

Example:

john@gmail.com

↓

e7f0d8a9....

Phone Number Protection

Phone numbers are masked.

Example:

9876543210

↓

XXXXX43210

Benefits:

- Protects customer privacy
 - Prevents exposure of sensitive information
 - Supports data governance requirements
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9. Data Enrichment

The cleaned transaction dataset is enriched using product master data.

Activities:

- Join transactions with product_details
- Resolve duplicate columns
- Validate product mappings
- Improve reporting quality

Benefits:

- Better category reporting
 - Improved consistency
 - Enhanced analytics capability
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10. Gold Layer Processing

Purpose:

Create an analytics-ready dataset.

Generated Features:

Revenue

Revenue = Price × Quantity

Profit

Profit = Revenue × 20%

Order Value

Revenue per transaction

Date Dimensions

- Year
- Quarter
- Month
- Day

Output:

data/gold/retail_curated.csv

11. KPI Definitions

The following KPIs were generated.

Total Revenue

Total sales generated across all transactions.

Total Orders

Total number of unique transactions.

Total Quantity Sold

Total quantity of products sold.

Total Profit

Estimated profit generated.

Revenue by Category

Sales contribution of each category.

Revenue by City

Sales contribution across cities.

12. Power BI Dashboard

The Gold dataset is used as the source for reporting.

Dashboard Components:

KPI Section

- Total Revenue
- Total Orders
- Total Quantity
- Total Profit

Revenue Analysis

Revenue Trend Over Time

Category Analysis

Revenue by Category

Regional Analysis

Revenue by City

Interactive Features

- Filtering
 - Cross Filtering
 - Drill Down Analysis
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13. Business Insights

1. Electronics contributes the highest revenue among all categories.
 2. Major metropolitan cities generate the largest share of sales.
 3. Revenue trends indicate seasonal fluctuations in customer purchases.
 4. Category-wise analysis helps identify high-performing products.
 5. Revenue concentration highlights key business markets.
 6. KPI monitoring enables data-driven decision making.
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14. Assumptions

1. Failed transactions are excluded from revenue calculations.
 2. Product master data is treated as the source of truth.
 3. Invalid quantities and prices are removed.
 4. Profit is estimated using a fixed business margin.
 5. Missing critical values are excluded from reporting.
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15. Challenges Encountered

- Missing values
- Duplicate transactions
- Data inconsistency
- PII protection requirements
- Date standardization
- Product standardization

These challenges were addressed through data validation and transformation logic.

16. Future Enhancements

- Azure Data Factory Integration
 - Azure Data Lake Storage
 - Apache Spark Processing
 - Incremental Loading
 - Workflow Scheduling using Airflow
 - Real-Time Reporting
 - Cloud Deployment
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17. Conclusion

This project successfully demonstrates an end-to-end Retail Sales Data Engineering Pipeline using Python and Power BI.

The solution performs ingestion, validation, cleaning, transformation, enrichment, KPI generation, and visualization while following industry-standard Medallion Architecture principles.

The final dashboard enables stakeholders to monitor business performance and make informed decisions using accurate and trusted data.